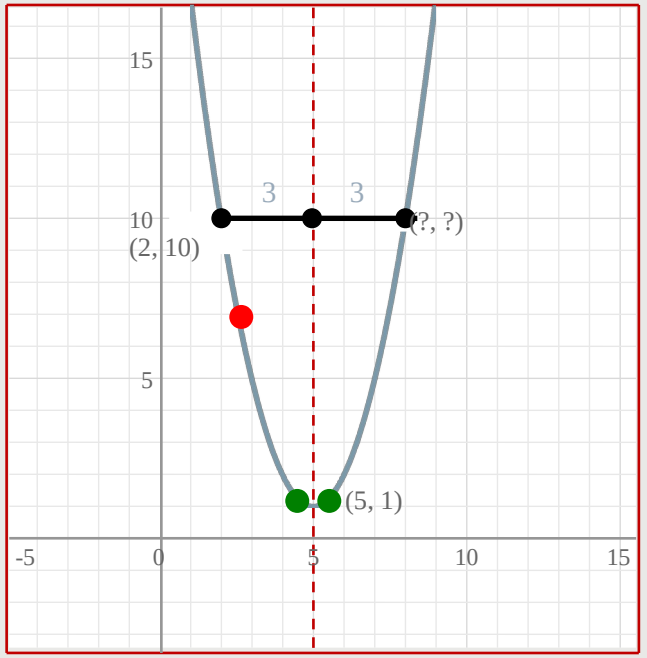
Convex Function, Learning Rate =0.3, Start at X=3

step	0.3
initial X	3
delta	0.1

$$f(x) = (x-5)^2 + 1$$



X is in between 4.8 and color red bold 5.1
f(x) is in between 1.04 and 1.01

Bouncing

x	3	3.3	3.6	3.9	4.2	4.5	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1
f(x)	5	3.89	2.96	2.21	1.64	1.25	1.04	1.01	1.04	1.01	1.04	1.01	1.04	1.01	1.04	1.01	1.04	1.01
x-delta	2.9	3.2	3.5	3.8	4.1	4.4	4.7	5	4.7	5	4.7	5	4.7	5	4.7	5	4.7	5
f(x-delta)	5.41	4.24	3.25	2.44	1.81	1.36	1.09	1	1.09	1	1.09	1	1.09	1	1.09	1	1.09	1
x+delta	3.1	3.4	3.7	4	4.3	4.6	4.9	5.2	4.9	5.2	4.9	5.2	4.9	5.2	4.9	5.2	4.9	5.2
f(x+delta)	4.61	3.56	2.69	2	1.49	1.16	1.01	1.04	1.01	1.04	1.01	1.04	1.01	1.04	1.01	1.04	1.01	1.04
slope	-8	-6.8	-5.6	-4.4	-3.2	-2	-0.8	0.4	-0.8	0.4	-0.8	0.4	-0.8	0.4	-0.8	0.4	-0.8	0.4
next x	3.3	3.6	3.9	4.2	4.5	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1	4.8	5.1	4.8